

Practicing Data Wrangling with Tree Equity Data

in-class worksheet viz

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Instructions

In this worksheet, you'll practice writing **ggplot R code** for creating data visualizations. Each question guides you through building layers of plots for the `ca_tes` dataset.

Don't stress yourself out over perfect syntax – the goal is to practice what you've learned about data wrangling so far.

Practice Questions

1. Tree Equity vs. Poverty

Is there a relationship between poverty (`pctpov`) and tree equity score (`tes`)?

```
ca_tes |>
  ___(x = ___, y = ___)) +
  ___(color = "green")
```

2. Distribution of tree equity scores

What is the distribution of tree equity scores (`tes`)?

```
___ |>
  ggplot(aes(___)) +
  ___(fill = "red")
```

3. Distribution of tree equity scores

What is the distribution of tree equity scores (`tes`) created by a boxplot?

```
___ |>
  ggplot(aes(___)) +
  ___(fill = "violet")
```

4. Tree equity vs. linguistic isolation

Is there a relationship between linguistic isolation(`lingnorm`) and `tes` in an area.

- Use `ggplot` to plot the relationship between `tes` and `lingnorm`
- Choose the best plot for the variable types
- Create fitting labels for your axes and plot title

5. Wrangle + plot

Recreate the plot above, but summarize the information **per county** before plotting. That is, **Los Angeles County** should now only be represented by one dot. Summarize both `lingnorm` and `tes` using the `mean()` function.

6. Wrangle + plot + label

Recreate the plot above, but now add labels to the plot to indicate the county of each dot. Use `geom_label()` or `geom_text()`. Try both! You'll need to use `label = county...` do you need `aes()`?

7. Wrangle + plot + label

There is an R package called **ggrepel**. It includes a geom called `geom_label_repel`. Try it! (To use a function from a package, you can write: `ggrepel::geom_label_repel()`.)

Hint: If R can't find the **ggrepel** package, you may need to type `install.packages("ggrepel")` in the Console down below.